Bracketing eigenvalues from a tension curve

The mathematics of `chladni`'s v1 bracket search

chladni development notes

Abstract

The method of particular solutions turns the Dirichlet eigenvalue problem on a planar domain into the problem of finding the local minima of a scalar *tension* curve $\sigma_1(\lambda)$, the smallest generalized singular value of a matrix pencil. This note develops the algorithm `chladni` uses for the first half of that search: given a window $[a, b]$ and a requested relative precision prec, produce a set of intervals each believed to contain one eigenvalue, and then drive each down to machine accuracy. Four ideas carry the design. (i) Near an eigenvalue σ_1 is a cone with a rounded bottom, whose two parameters — a floor ε belonging to the *basis* and a wall slope c belonging to the *domain* — are independent, and whose ratio ε/c is the resolution limit. (ii) The wall slope scales as $c(\lambda) = C/\sqrt{\lambda}$, which makes $k = \sqrt{\lambda}$ the *unique* power of λ in which one Lipschitz bound serves a whole window; this change of variable is worth more than every algorithmic change in the design combined. (iii) A Lipschitz bound converts samples into intervals *certified to contain no eigenvalue*, and the cost of certifying a cone's wall is a geometric ladder whose ratio depends only on the safety factor — which fixes the cost of the method and retires a family of attempts to make it cheaper. (iv) One evaluation returns the whole singular-value vector, so σ_2 is free, and σ_2/c is a distance meter that is accurate exactly where the decision is hard. We derive each, state honestly what is proved and what is only measured, and record what changed when the algorithm was ported from the repository it was designed in.

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1 What the bracketing stage is asked for

1.1 From the pencil to a scalar curve

Let $\Omega \subset \mathbb{R}^2$ be a bounded planar domain and consider the Dirichlet eigenvalue problem $-\Delta u = \lambda u$ in Ω, $u = 0$ on $\partial\Omega$. The method of particular solutions (MPS) builds, at each trial value λ, a basis $\psi_1(\cdot;\lambda), \ldots, \psi_n(\cdot;\lambda)$ of functions that solve the Helmholtz equation $\Delta\psi + \lambda\psi = 0$ in Ω *exactly*.

Every member of the span is then an exact solution of the PDE, and only the boundary condition remains to be satisfied. Sampling the boundary at m_B collocation points and the interior at m_I points gives two matrices,

$$(A_B)_{ij} = \psi_j(x_i^B), \qquad (A_I)_{ij} = \psi_j(x_i^I),$$

and the question "is λ an eigenvalue?" becomes "is there a coefficient vector making the boundary trace small without making the function small everywhere?" That is measured by the *tension*

$$\sigma_1(\lambda) \;=\; \min_{c \neq 0} \; \frac{\|A_B(\lambda)\, c\|}{\|A_I(\lambda)\, c\|}, \tag{1}$$

the smallest generalized singular value of the pencil $\big(A_B(\lambda), A_I(\lambda)\big)$. The interior block is what stops the minimizer from returning the zero function; it is a normalization, not a quadrature, and its only job is to keep $\|A_I c\|$ away from zero.

Three properties of σ_1 shape everything below.

1. **It dips to a floor at each eigenvalue and is $O(1)$ elsewhere.** The elevated region between dips is the *plateau*, and its height is the natural scale against which smallness is judged.

2. **One evaluation returns the whole ascending vector** $\sigma_1 \leq \sigma_2 \leq \cdots$. An eigenvalue of multiplicity m drives m of them to the floor together, and reading σ_2 costs nothing extra. Section 6 spends this.

3. **Evaluations are expensive.** One costs a QR, an SVD and a generalized SVD — 5–70 ms depending on basis size. Cost is therefore counted in evaluations and in nothing else, and every design decision below is ultimately a decision about how many to spend.

1.2 The output contract, and its two metrics

Definition 1 (The bracketing problem)**.** Given the tension function, a window $[a, b]$ and a requested relative precision prec, return a set of non-overlapping intervals, each of which the method *believes* contains an eigenvalue.

Two properties of the returned set are hard requirements.

Coverage. Every eigenvalue in $[a, b]$ lies in some returned interval. A lost eigenvalue cannot be recovered downstream by any amount of refinement, so this is the metric that governs.

Purity. No returned interval contains eigenvalues spanning more than $\text{prec} \cdot \lambda$. An interval holding two eigenvalues the user asked to have separated is a wrong answer, not an imprecise one.

Everything else — how many intervals, how many of them are empty, what it all cost — is reported and traded rather than required.

Remark 1 (The stage must commit)*.* The returned intervals are the ones the method believes hold an eigenvalue, *not* a subdivision of $[a, b]$. The distinction is not pedantic: a method that partitions the window achieves perfect coverage for free and without evaluating anything, while a method that commits covers a small fraction of the window and has to earn it. Without this rule the two are not comparable at all.

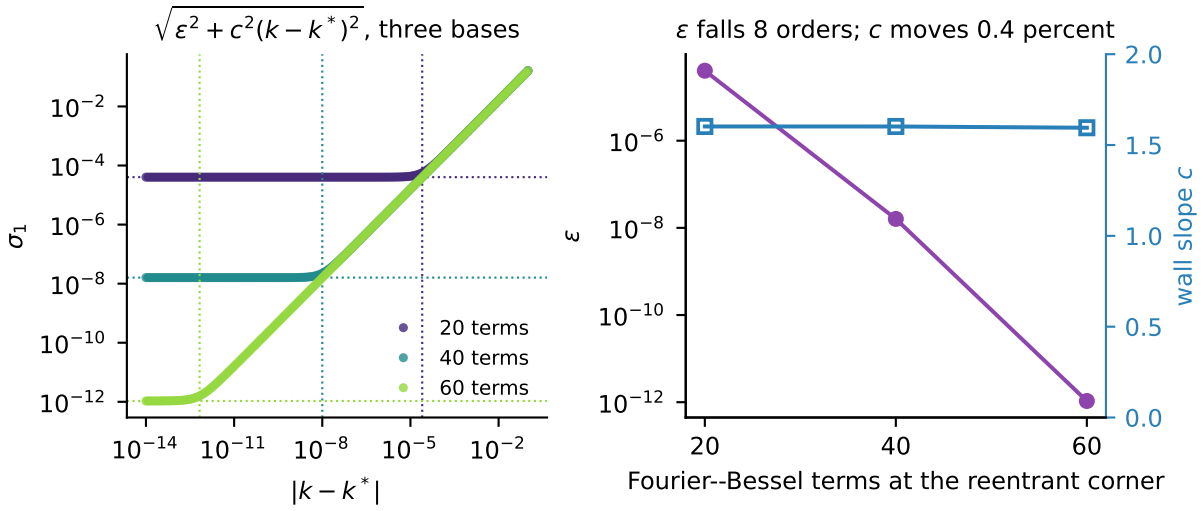


Figure 1: The cone model on the L-shape, at the first eigenvalue, with a Fourier–Bessel basis at the reentrant corner. *Left:* measured σ_1 (points) against (2) (lines) for three basis sizes; dotted lines mark ε and ε/c. The three curves share a wall and differ only in where they bottom out. *Right:* the same three fits. Only ε responds to the basis.

Note what purity does *not* constrain: the *width* of an interval holding a single eigenvalue. Such an interval spans zero eigenvalues however wide it is. This observation is worth remembering; it is the whole of §7.2's argument for stopping early.

2 The local model, and what its two parameters mean

Near a simple eigenvalue λ^*, the tension is well described by

$$\sigma_1(\lambda) \;\approx\; \sqrt{\varepsilon^2 + c^2\,(\lambda - \lambda^*)^2}, \tag{2}$$

a cone of slope c with a rounded bottom of depth ε. On the four domains where it has been checked — rectangle, disk, equilateral triangle, L-shape — (2) fits the far field to a median relative residual of 1–3%, failing only where the floor comes close to the plateau, that is, where the basis cannot represent the eigenfunction at all.

The two parameters are independent, and *which is which* matters more than either value.

ε **is the floor.** It is σ_1 at the eigenvalue: the residual of the best eigenfunction the finite basis can represent. It is dimensionless, and it is a property of **the basis**.

c **is the wall slope.** It is $d\sigma_1/d\lambda$ away from the eigenvalue, it carries units, and it is a property of **the domain**.

Figure 1 is the evidence. On the L-shape with a Fourier–Bessel basis at the reentrant corner, taking 20, 40 and 60 terms moves the floor from 4.0×10^{-5} to 1.1×10^{-12} — seven orders — while the wall slope moves from 1.6022 to 1.5951, four parts in a thousand. Enriching the basis deepens the well; it does not change its walls.

2.1 The resolution limit

Setting $\lambda - \lambda^* = \varepsilon/c$ in (2) gives $\sigma_1 = \sqrt{2}\,\varepsilon$: the ratio

$$\frac{\varepsilon}{c} \tag{3}$$

is the scale at which the cone stops looking like a wall and starts looking like its own rounded bottom. Two eigenvalues closer together than ε/c have overlapping bottoms and cannot be separated by *any* method reading this curve, however many evaluations it spends. Since c is fixed by the geometry, ε alone sets the limit — which is the concrete form of the statement that *the search is bounded by the basis*. It is the basis planner's job to make ε small enough; it is the search's job to find everything that is separated by more than about $20\,\varepsilon/c$, which is the quantitative form of the contract between the two layers.

Remark 2 (No absolute threshold on σ can serve as a test). The floor is not a single number even within one scan, and the reason is worth stating precisely because a loose version of it is easy to repeat. At a corner of interior angle ω the local expansion of an eigenfunction runs in

$$r^{\,l\pi/\omega}\,\sin\!\big(l\pi\theta/\omega\big), \qquad l = 1, 2, \ldots,$$

whose exponents are fractional unless ω/π divides l. Every Dirichlet eigenfunction *vanishes* at the corner — that is the boundary condition, and it distinguishes nothing. What distinguishes one mode from another is whether the *coefficients of the fractional-exponent terms* vanish. A mode with no singular part is analytic there and a finite basis represents it almost exactly; its neighbours carry the singularity and are represented far less well. On the L-shape, $\omega = 3\pi/2$ gives exponents $r^{2l/3}$, smooth only when $3 \mid l$; the mode at $\lambda = 2\pi^2$ is the unit square's fundamental $\sin(\pi x)\sin(\pi y)$ tiled across the three cells, which vanishes on the whole integer grid and is entire.

Every threshold in the algorithm is therefore stated as a fraction of the *measured plateau*, never as an absolute tension.

The measurement behind this rule has itself been corrected, and the correction is instructive. An early measurement recorded a five-order floor spread on the L-shape — 3.1×10^{-13} at one level against 5.5×10^{-8} at a neighbour — and explained it structurally. With a recipe-grade basis (80 pure Fourier–Bessel terms at the reentrant corner) the same levels read within a factor of 4.2 of one another, and the level previously at 5.5×10^{-8} reads 5.3×10^{-15}. So the five orders were an artifact of an inadequate basis. The *design rule* survives either way, because a factor of four still defeats any fixed constant and a domain whose basis cannot be made adequate will still show the large spread; what does not survive is the claim that the L-shape is intrinsically such a domain.

3 Why the wavenumber, and why uniquely

This section contains the shortest derivation in the document and the one that decides the most.

The cover phase of §5 rests on a single Lipschitz bound L used across a whole window. Such a bound is conservative by the ratio of the largest local slope in the window to the smallest, and it is paid *everywhere*. Reducing that ratio is therefore the largest available saving, and it is available before any algorithm has been chosen.

Measurement supplies the scaling: on the family where it was first taken,

$$c(\lambda) \;=\; \frac{C}{\sqrt{\lambda}}, \qquad C = 0.27 \pm 8\%, \tag{4}$$

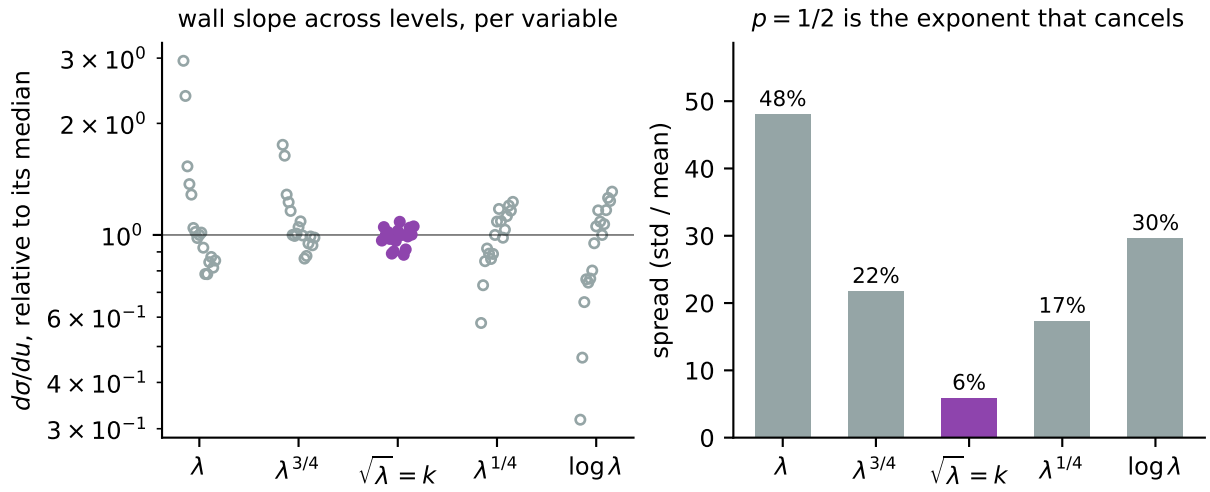


Figure 2: The wall slope across 17 isolated levels of a $1 \times \varphi^{-1}$ rectangle, under five changes of variable. *Left:* each level's slope relative to the median for that variable. *Right:* the spread. The exponent that cancels is the one the data pick out.

so c varies by a factor of 4.81 across $\lambda \in [10, 400]$. Now search not in λ but in $u = \lambda^p$ for some power p. By the chain rule,

$$\frac{d\sigma}{du} \;=\; c(\lambda)\,\frac{d\lambda}{du} \;=\; \frac{C}{\sqrt{\lambda}} \cdot \frac{\lambda^{1-p}}{p} \;=\; \frac{C}{p}\,\lambda^{1/2-p}, \tag{5}$$

which is constant in λ for $p = 1/2$ and for no other power. This is not a fit; it is the exponent that cancels.

Proposition 1. *Under the scaling (4), $u = \lambda^p$ makes the wall slope independent of the level if and only if $p = 1/2$; that is, if and only if u is the Helmholtz wavenumber $k = \sqrt{\lambda}$.*

That k is also the physical variable — the wavenumber at which the basis functions are built — is the reason this is expected to transfer to other domains, rather than being a property of the family it was measured on.

Figure 2 confirms it on a different rectangle, in a different implementation, from the one where (4) was measured. Over 17 isolated levels in $\lambda \in [10, 520]$, the spread of $d\sigma/du$ (standard deviation over mean) is 48% in λ, 22% in $\lambda^{3/4}$, 17% in $\lambda^{1/4}$, 30% in $\log \lambda$, and **5.9% in** k. The originating measurement reported 57%, 27%, 17%, 32% and 5%. Two independent implementations, two different rectangles, the same answer.

Remark 3 (How much this is worth). In the design's own version history this change of variable accounts for a factor of 1.8 out of a total 2.6 improvement — more than every algorithmic change in the method combined. It is worth stating plainly, because it is easy to read as a numerical detail and it is not one.

3.1 Two consequences of working in a transformed variable

The precision factor. Purity is judged on $d\lambda$ against $\mathrm{prec} \cdot \lambda$, but the search contracts brackets in u. A bracket $[u_1, u_2]$ maps to $d\lambda = (d\lambda/du)\,du$, so for $u = \lambda^p$ the relative precision that must

be met in u is

$$\mathrm{prec}_u \;=\; p \cdot \mathrm{prec}, \tag{6}$$

a *half* for the wavenumber. Evaluated at the upper end of the bracket, so conservative.

The ulp slack. The round trip $k \mapsto k^2$ is not exact. A bracket that correctly fences k^* can fail to fence $\lambda^* = (k^*)^2$ after squaring, because an endpoint rounds the wrong way; this was observed as a coverage loss of exactly one unit in the last place. Every returned endpoint is therefore widened by about four ulps, and brackets that abut after widening are merged.

And the refiner wants the other variable. §9 shows that the quadratic fit the refinement performs is markedly better in λ than in k. The two stages want two variables and they get them: **cover in k, refine in λ.** This is not a compromise between two measurements but the conclusion of both, and it is worth noticing that the reasons are genuinely different — the scan cares about the *slope*, which is flat in k, and the refiner cares about the *curvature*, which is better behaved in λ.

4 Seeding the scan, and estimating the slope

4.1 Weyl sizes the grid and says nothing else

Weyl's law with the perimeter correction,

$$N(\lambda) \;\sim\; \frac{|\Omega|\;\lambda}{4\pi} \;-\; \frac{|\partial\Omega|\,\sqrt{\lambda}}{4\pi}, \tag{7}$$

predicts the number of eigenvalues in a window to 1–11% on the families tested. It is used for exactly one thing: to set the starting sample density, at 1.5 samples per expected gap. It says nothing whatever about *where* anything is, and must never be used as though it did.

The starting density barely matters, because the doubling below adapts: 0.5, 1.0 and 1.5 samples per gap all cost the same. The larger is chosen for margin.

4.2 Nested doubling, and why $n \to 2n$ is wrong

The seed then doubles, inserting midpoints so that $n \to 2n-1$ and each grid *contains* the previous one exactly, until the largest observed secant slope stops growing by more than 10%.

Exact nesting is not tidiness. Under $n \to 2n$ the new points land at irrational offsets from the old ones, and some of them land within an ulp of an old point. A secant computed across such a pair measures rounding, not slope: one such estimate reached 7.3×10^{10} against a true slope of 0.6. Since the slope model is monotone non-decreasing (§4.4), a model that once believes that can never recover. For the same reason, pairs closer than 10^{-9} of the sampled span are excluded from the secant estimate altogether. Midpoint insertion has a second benefit: the old points are reused bit-identically, so a doubling costs only the genuinely new samples.

4.3 The bootstrap hazard runs opposite to intuition

It is natural to worry that an *over*estimated Lipschitz constant will make the search expensive. The dangerous error is the other one. As §5 will show, a sample of value v excludes a neighbourhood

of radius v/L; an *under*estimated L makes those radii too **large**, so a search that has not yet seen the true slope excludes everything and terminates having found nothing — silently, with no error at all. Measured, the largest observed secant as a function of seed density, against a true slope of 0.0656:

n	8	16	32	64	128	256
largest secant	0.0046	0.0128	0.0386	0.0599	0.0647	0.0652
as a fraction of the truth	7%	20%	59%	91%	99%	99%

At $n = 8$ the exclusion radii would be fourteen times too large.

Two things defuse this. First, the doubling itself: the estimate is not used until it has stopped growing. Second, and more important, *exclusions are recomputed from the current L on every pass*, so a subsequent growth in L revives whatever that growth un-excludes. An underestimate therefore costs extra work rather than causing a silent miss.

Remark 4 (The bootstrap is the falsification test). An estimate that never settles is not a nuisance to be worked around: it is the evidence that nothing bounds the local slope on this curve, at any scale sampled. That is a legitimate ground for the method to decline (§10.3), and it is the only such evidence available.

4.4 The slope model, and what it is honestly worth

The model is

$$L \;=\; s \cdot \max_i \left| \frac{v_{i+1} - v_i}{x_{i+1} - x_i} \right|, \qquad s = 1.5, \tag{8}$$

monotone non-decreasing over the run. The honesty clause must be stated before anything is built on it.

Proposition 2 (There is no certificate here)**.** *No finite sample set bounds a Lipschitz constant from above. The largest observed secant is a* lower *bound on* $\sup |f'|$*, and multiplying it by a safety factor produces a number with no formal status.*

This is not a defect that better engineering removes; it is a property of the information available. The consequence is that every completeness claim the method makes is *conditional on* L, and must be reported as such (§10).

Two measurements set $s = 1.5$, and they pull in opposite directions.

- **The bound is nearly tight, so it cannot be shaved.** On the window where it was measured the largest observed secant reaches the true nearest eigenvalue exactly. Using the 99th-percentile slope instead of the maximum — a 1.4% reduction — wrongly excluded 13 of 600 samples. Coverage is lost by very small underestimates.
- **Margin is cheap in the seed.** Raising s from 1.2 to 2.0 cost 40% more evaluations for identical coverage.

Finally, distinguish a *surprise* growth — L rising after exclusions have already relied on it — from mere convergence of the estimate. A creep from 0.0647 to 0.0656 is the estimate settling. Only surprises are counted, and a run with many of them deserves a rerun at a larger s; it is not, by itself, wrong.

4.5 An aside: could the scan have found it at all?

Sizing the scan is a *precondition*, not a diagnostic, and `chladni` ships a closed-form check for it. Model the dip as a clipped V, $\sigma \approx \min(\sigma_{\text{base}}, C_k\,|k - k_j|)$, where σ_{base} is the plateau. Its visible width is $w_k = 2q\,\sigma_{\text{base}}/C_k$ for a visibility fraction q, and demanding that a scan of stride Δk resolve it, $w_k \geq \Delta k$, gives

$$\sigma_{\text{base}} \;\geq\; \frac{\kappa}{2q} \cdot \frac{|\Omega|}{|\partial\Omega|} \cdot \Delta k, \qquad \kappa \;=\; \frac{C_k}{|\Omega| \,/\, |\partial\Omega|}. \tag{9}$$

The $|\Omega| \,/\, |\partial\Omega|$ scaling is confirmed exactly: the same $3:1$ rectangle at scales 0.5, 1, 2 gives $C_k/(A/P) = 2.4600,\ 2.4600,\ 2.4598$, while C_k/A varies by a factor of four over the same three. The dimensionless κ transferred between two independent implementations within one standard deviation, 2.591 ± 0.226 against 2.384 ± 0.231.

5 Cover by exclusion

5.1 The Piyavskii bound

Suppose f is L-Lipschitz and we know $f(x_i) = v_i$ and $f(x_j) = v_j$ at two adjacent samples, $x_i < x_j$, $h = x_j - x_i$. Then for every $x \in [x_i, x_j]$,

$$f(x) \;\geq\; \max\big\{\, v_i - L(x - x_i),\;\; v_j - L(x_j - x) \,\big\}, \tag{10}$$

the pointwise maximum of two cones of slope L dropping from the two samples. The right-hand side is minimized where the two branches cross, giving

Lemma 1 (The envelope and its minimizer)**.** *The bound* (10) *attains its minimum*

$$\hat{s} \;=\; \frac{v_i + v_j - Lh}{2} \qquad \textit{at} \qquad \hat{x} \;=\; \frac{x_i + x_j}{2} + \frac{v_i - v_j}{2L}. \tag{11}$$

Proof. The two branches are affine with slopes $-L$ and $+L$; their maximum is minimized at their crossing, $v_i - L(x - x_i) = v_j - L(x_j - x)$, which rearranges to $\hat{x}$ above and gives the value $\hat{s}$ by substitution. □

Two consequences, and the method is built from them.

The exclusion certificate. If $\hat{s} \geq 0$ then $f > 0$ throughout (x_i, x_j), so no eigenvalue lies between the two samples. The cell is *certified empty* — conditional on L, per Proposition 2.

Where to look next. $\hat{x}$ is the point attaining the bound: the most informative place in the cell, and where the search samples.

Cells not excluded are *live*, and contiguous live cells are merged into *regions*. Merging matters: the two cells either side of an eigenvalue are both live, and reported separately they would look like two candidates where there is one.

Figure 3 shows the construction on a real tension curve.

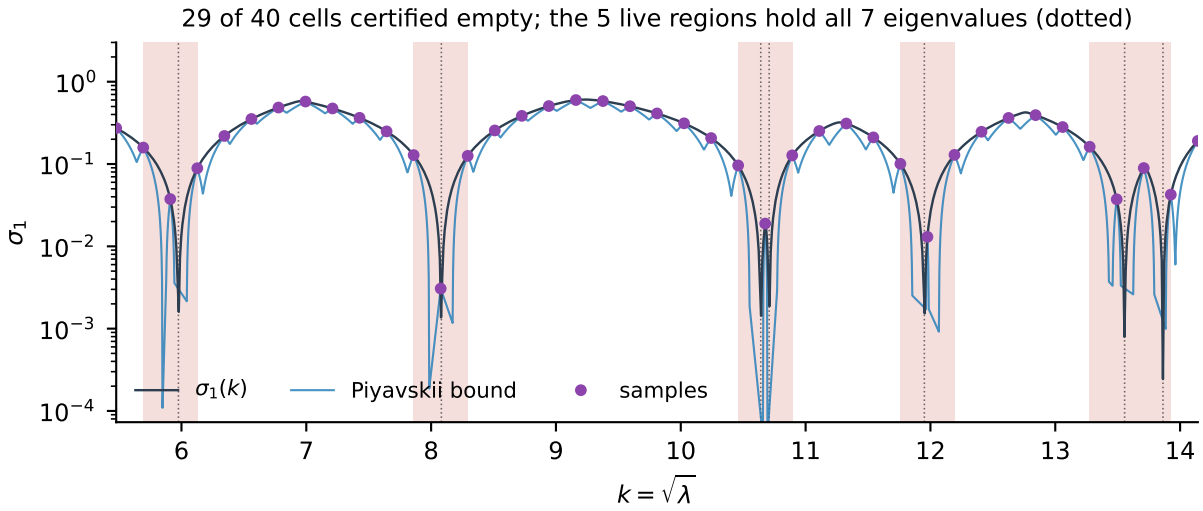


Figure 3: Cover by exclusion on a $1 \times \varphi^{-1}$ rectangle, $\lambda \in [30, 200]$, from 41 uniform samples with $L = 1.5\times$ the largest observed secant. The sawtooth is the Piyavskii bound (10), drawn only where it is positive. 29 of 40 cells are certified empty; the five surviving live regions (shaded) contain all seven eigenvalues (dotted). Two of them hold a pair, which is what §6 exists to detect.

Remark 5 (The zero threshold, and why it does not need to be raised). The test $\hat{s} \geq 0$ presumes σ_1 reaches zero at an eigenvalue, and it does not: it bottoms at $\varepsilon > 0$. So as the spacing shrinks the bound rises toward a positive value and a sufficiently finely sampled well could, in principle, exclude *itself*. This is a reasonable objection and the answer is structural rather than a tuned constant: the cover loop stops on purity-by-width long before that regime is reached, so the zero threshold never bites. Replacing it with a positive threshold was tried and broke every case — 0 of 66.

5.2 The cost law

What does it cost to certify a cone's wall as empty? Take two samples on the same wall at distances $u_a > u_b$ from the eigenvalue, so their values are $c\,u_a$ and $c\,u_b$, and take $L = s\,c$. By (11) the cell between them is excluded when

$$\frac{c(u_a + u_b)}{2} - \frac{s\,c\,(u_a - u_b)}{2} \;\geq\; 0 \quad\Longleftrightarrow\quad (1 - s)\,u_a + (1 + s)\,u_b \;\geq\; 0,$$

that is, when

$$\frac{u_b}{u_a} \;\geq\; \frac{s - 1}{s + 1}. \tag{12}$$

Proposition 3 (The ladder)**.** *The condition* (12) *is independent of* c*. A cone's wall is therefore covered by a* geometric ladder *of samples with ratio at most*

$$r(s) \;=\; \frac{s + 1}{s - 1},$$

and covering a range of scales R *costs* $\log R / \log r(s)$ *samples per side.*

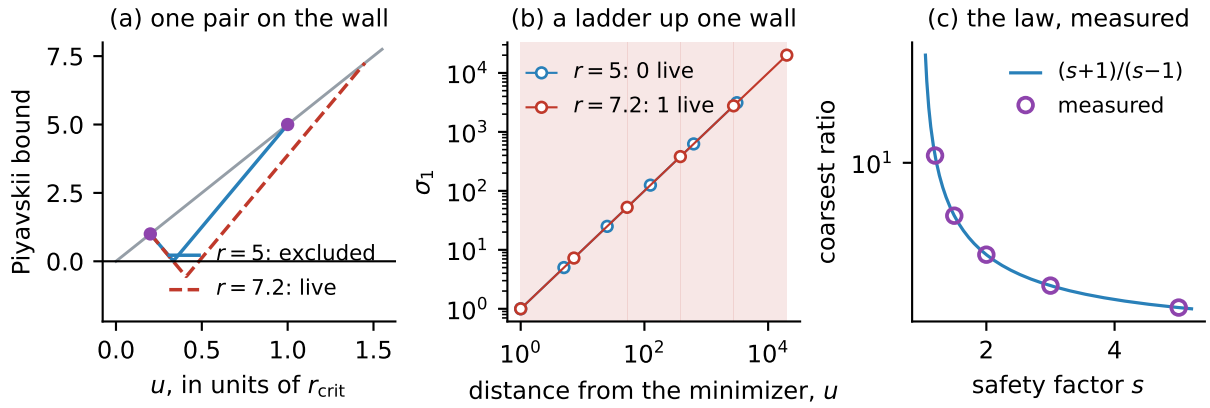


Figure 4: *(a)* One pair on a wall. At the critical ratio the Piyavskii bound just touches zero; past it the bound dips below and the cell stays live. *(b)* A ladder up one wall: at $r = 5$ every cell is excluded, at $r = 7.2$ none is. *(c)* The ratio measured by bisection against `live_regions`, against $(s+1)/(s-1)$.

Two things about this deserve emphasis. It is independent of the wall slope, so it holds uniformly over the window and over domains. And it is exact: Figure 4 recovers $r(s)$ to five figures by bisecting on the ratio against the exclusion code itself.

Remark 6 (A better prediction cannot help the cover phase). Proposition 3 retires a family of ideas. Placing a sample *exactly on* an eigenvalue leaves the live region completely unchanged, and so does placing two that straddle it to within 10^{-9}: what is live is the *wall*, and the wall yields to the ladder or to nothing. Six separate attempts to make cover cheaper by sampling more cleverly all failed for this one reason, and a seventh would too. Cleverness about *where the eigenvalue is* belongs in the locator (§7), not here.

Remark 7 (A larger safety factor is a finer ladder). Raising s does not merely buy margin; it costs $1/\log r(s)$ per side, so the cover phase's cost grows as s rises. That is why the tension in §4.4 between "the bound is nearly tight" and "margin is cheap" is a real trade and not a free choice.

5.3 Why adapt at all

A uniform grid must resolve the tightest pair it might meet. A pair at separation d is resolved once the sampling near it is finer than about $d/3$, so a uniform grid over $[a, b]$ needs $n \sim 3(b-a)/d$ samples *everywhere*, while exclusion refines only where σ_1 is small and costs $O(\log(1/d))$. Measured on an interval of width 2:

d	uniform grid	exclusion	ratio
3×10^{-3}	2048	47	43×
10^{-4}	$\sim 6 \times 10^4$	57	1053×
10^{-6}	$\sim 6 \times 10^6$	71	8.5×10^4
10^{-10}	$\sim 6 \times 10^{10}$	97	6×10^8

The comparison matters because d is *not known in advance*. A black-box solver cannot tune its grid to the tightest pair it will meet, so the uniform-grid column is a cost it would have to pay everywhere and forever, against a pair it may or may not have.

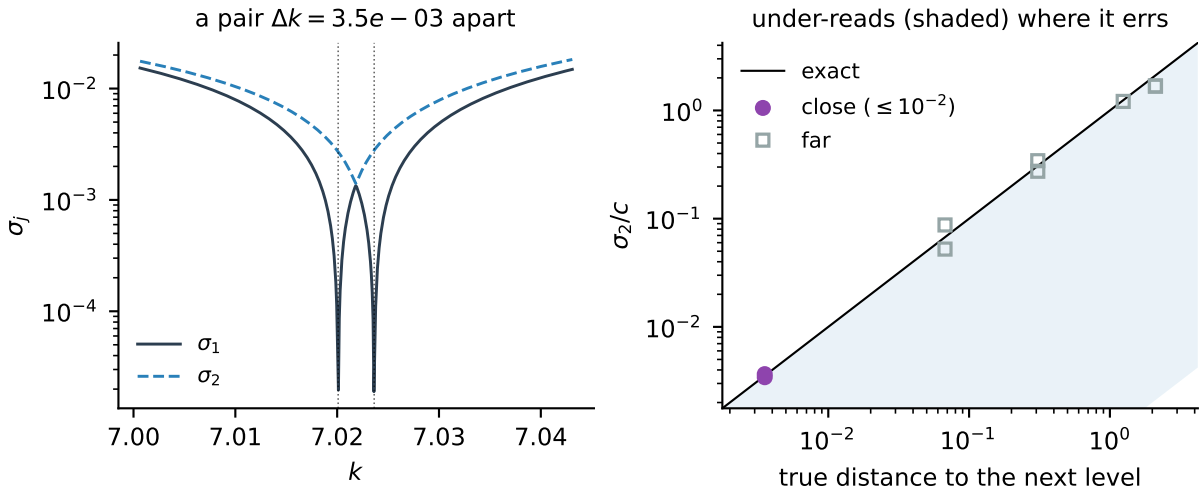


Figure 5: *Left:* σ_1 and σ_2 across a resolved pair on a $(1+\delta) \times 1$ rectangle. σ_2 has a single minimum, at the crossing point. *Right:* the reading σ_2/c against the true distance to the next level, over nine levels of two pencils, computed by the searcher's own `isolation` routine. The shaded half-plane is where the meter under-reads.

6 The isolation gate: what σ_2 buys

6.1 Why a gate is needed

Once a region is believed to hold one eigenvalue, a fast locator can drive it down cheaply. But a locator that converges to a single minimizer will happily converge to *one member of a close pair* and leave the other outside the bracket it returns. This is not hypothetical: it is exactly how a grid-then-bisect baseline falls from coverage 1.0 to 0.577. The fast path must therefore be gated, and the gate must be right.

6.2 The signal, and why it is conservative by construction

One evaluation returns the whole ascending vector, so σ_2 is free. Near an eigenvalue, σ_2 is the second branch of the pencil — which, at a simple level, is the *next* eigenvalue's cone — so

$$\frac{\sigma_2(x)}{c} \;\approx\; \text{distance from } x \text{ to the next eigenvalue.} \tag{13}$$

A region is certified to hold one eigenvalue when

$$\text{neighbour} \;>\; \gamma \cdot (\text{region width}), \qquad \gamma = 1.5. \tag{14}$$

The estimator's error structure is what makes this safe, and it is worth stating carefully. Measured against the true gap, (13) reads at ratio 0.97–1.03 when the neighbour is close, and 0.78–1.30 when it is far (Figure 5). It is accurate *precisely where the decision is hard* — when a neighbour is close enough to be a purity risk — and where it is inaccurate it generally *under*-reads the gap. A gate phrased as "is the neighbour further away than this region is wide?" therefore errs toward refusing to certify. It is conservative by construction, not by tuning.

6.3 Estimating c without cancellation

The gate needs c, and it reads it off the descent triple. If sampled points lo < mid < hi straddle the minimizer then, on a cone, $v_{\text{lo}} = c(\lambda^* - \text{lo})$ and $v_{\text{hi}} = c(\text{hi} - \lambda^*)$, so adding,

$$c \;=\; \frac{v_{\text{lo}} + v_{\text{hi}}}{\text{hi} - \text{lo}}. \tag{15}$$

No subtraction of nearly equal quantities appears anywhere, and the estimate does not depend on where λ^* is within the triple.

6.4 Four rules, each of which cost a measured failure

1. **Judge the region, not the triple inside it.** The region is what gets handed to the locator and can be several times wider than the certificate triple. A pair 8.4×10^{-4} apart was certified inside a region 1.24×10^{-3} wide, and then collapsed onto one member.

2. **c must be an independent *upper* bound**, taken as $\max(c_{\text{local}}, L)$. Where a region holds *two* eigenvalues its flanks span both cones, so hi − lo in (15) is too wide, c comes out too small, the neighbour appears further away than it is, and the gate certifies precisely what it must not. Trusting the local estimate alone cost 3 of 26 eigenvalues.

   This is worth dwelling on, because the instinct is the reverse. Replacing the crude global L with a *learned* wall slope — an obviously better estimate — loses 12 of 19 cases. A smaller slope makes the estimated neighbour distance larger, which pushes the gate toward certifying. The global bound's apparent excess conservatism *is* the safety margin.

3. **Silence is not permission.** If no descent certificate exists in the region there is no decision to be had, and the region stays on the slow path. "I cannot tell" must not be recorded as "yes".

4. **Do not read the level's multiplicity off the σ vector here.** Counting branches at the floor and reading the first one that is not is correct *in principle* — at a multiplicity-2 level, σ_2 is that level's own second branch, and the next distinct level is reported by σ_3 — and it saves 40% on exactly degenerate cases. It is also ambiguous at range by construction: approaching an unresolved pair of separation d from distance u,

   $$\frac{\sigma_2}{\sigma_1} \;=\; 1 + \frac{d}{u}, \tag{16}$$

   which is 1.0006 at the start of a cover pass. No threshold separates that from a genuine multiplicity-2 level. Swept over two windows, the failure count is **not monotone** in the threshold — $0, 1, 0, 1, 2$ failures at $1.0, 1.05, 1.2, 1.5, 2.0$ — and a parameter whose safety alternates as it is tightened has no safe setting, only settings that pass the cases tried.

Remark 8 (The reading is not wrong; its placement was). Read *after* location, against a deliberately wide bracket, the same quantity is unambiguous: 14 levels, 14 correct multiplicity readings, at zero additional evaluations, because every point needed is already in the cache. Within a multiplet the ratio σ_{j+1}/σ_j runs 1.0–1.7 and across the gap 10^{11}–10^{15}, so a threshold of 10 is not a delicate parameter at all. What (16) forbids is reading it *at range*, which is the only place the bracketer could.

7 Locate: predict, then straddle

7.1 The prediction is exact, and it is not enough

Under the cone model (2),

$$\sigma_1^2 \;=\; \varepsilon^2 + c^2(\lambda - \lambda^*)^2 \tag{17}$$

is *exactly* quadratic. Three points therefore determine the minimizer exactly rather than approximately, and the residual error comes only from the two branches meeting at the bottom having slightly different slopes. This is also far better conditioned than the obvious alternative of extrapolating a wall to zero, which subtracts nearly equal values.

Squaring is not a numerical sin here; see Remark 13.

Now the insight that makes the locator work.

Remark 9 (Prediction alone does not tighten a bracket). A bracket needs sampled points on *both* sides of the minimizer, so its width is set by whichever side of the prediction has no sample near it. The prediction error has a *systematic sign*, because the two branches have slightly different slopes. So every scheme that samples *at* the prediction converges from one side only, and the far flank never moves.

The fix is to place the sample a fixed fraction of the bracket to one side of the prediction, on whichever side is not already fenced. Each round of two such samples shrinks the bracket to about $2\times$ the straddle fraction. That fraction is bounded below by the prediction error — an offset smaller than the error lands on the wrong side and the triple stops being a descent — and above by the contraction it buys.

Figure 6 measures both bounds. The parabola's vertex lands within 1.2×10^{-4} of the truth at the median and 3.1×10^{-3} at the 90th percentile, as fractions of the bracket; the cost rises steadily with the offset above that. The shipped value 0.02 sits about six times above the measured 90th percentile of the prediction error, and costs 14 evaluations against 12 at 0.01 on this pencil.

Remark 10 (Three designs that fail, each plausible). Plain interpolation — sampling at the prediction — stalled with the bracket stuck. Bisecting the lopsided side cured the stall but cost no less than exclusion. Mirroring the near sample across the prediction oscillated without converging. Each is a natural thing to try, and each is worth knowing has been tried.

7.2 What licenses emitting a bracket

Every endpoint must be sampled, and carry a descent certificate. For three sampled points $p < q < r$ with $f(p) > f(q) < f(r)$, continuity alone gives a local minimum in (p, r). Nothing weaker may define an endpoint. In particular an *extrapolated* root may not: two-point secant roots fitted from opposite walls of a close pair were measured to **cross** — the left root landing to the right of the right root — yielding an inverted bracket, or a sliver containing nothing. Extrapolation may choose *where to sample*; it may not name an endpoint.

Two details of the certificate are easy to get wrong, and were.

- **Anchor on the region's lowest sample**, not on the deepest-looking triple. This is what makes the bracket tighten monotonically: every new sample nearer the bottom becomes the new centre, and its immediate neighbours are necessarily closer than the old ones. Selecting by depth instead stalls, because depth tends to 1 as the flanks approach the middle.

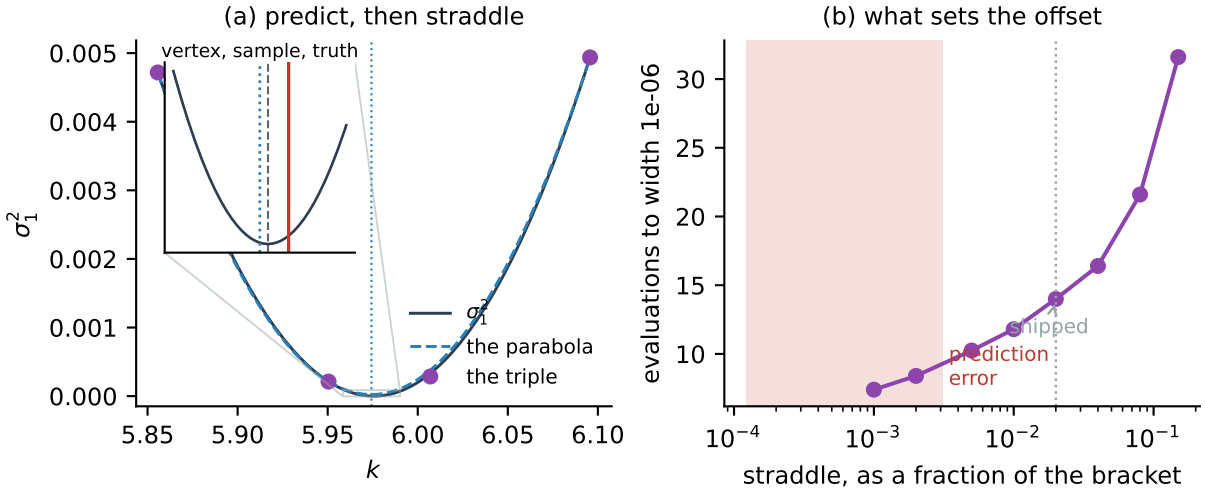


Figure 6: *(a)* One locator step. σ_1^2 and the parabola through the triple are indistinguishable at full scale; the inset zooms on the bottom, where the predicted vertex (dotted), the sample placed 2% to its unfenced side (solid) and the true minimizer (dashed) separate. *(b)* Evaluations to contract a bracket to width 10^{-6}, averaged over five isolated levels, against the straddle fraction. The shaded band is the measured prediction error over the same runs, median to 90th percentile: below it the offset is inside the noise of the prediction.

- **Immediate neighbours only.** A descent read off non-adjacent samples would skip whatever lies between them, and could straddle two cones with a barrier in between.

Depth, normalized by the further flank. To decide whether a region holds a minimizer at all, the middle's value is compared to its flanks. The normalization must be by the *larger* of the two flanking values. Using the nearer one systematically rejects real eigenvalues: when a cone bottoms out near the middle of a sampling cell its two flanking samples are almost equidistant, so their values are almost equal and the ratio approaches 1 however deep the cone actually is. That cost 27% of coverage — 19 of 26 eigenvalues instead of 26 — *at every grid density*, because refining the grid only moves the problem to a different cell. Against the further flank, a mid-cell cone scores about 0.33 while plateau ripple still scores about 0.98, which is the separation the ratio is for. The test is asked *once* per region, and never to choose among certificates: depth necessarily tends to 1 as a bracket tightens, so a locator that filtered on it would stall partway down.

7.3 Stop at isolation

Recall from §1 that purity does not constrain the width of a bracket holding one eigenvalue. Refining past isolation therefore buys nothing the contract asks for, and the refiner of §9 reaches machine precision from a wide bracket in about seven evaluations. So the bracket is handed on once the neighbour is four bracket-widths clear.

State plainly what this gives up. A bracket narrower than $\text{prec} \cdot \lambda$ cannot be impure *however wrong the gate was*; that unconditional guarantee is traded away, and purity then rests on §6. This is the change in the design most in need of confirmation on domains beyond the one it was measured on.

Remark 11 (prec is a stopping rule, not a merge licence). The search never reasons about whether merging two nearby eigenvalues into one bracket is *permitted*. The economics are entirely one-sided: merging saves the refinement of one bracket, certifying that a merge is legal means resolving both eigenvalues anyway, and being wrong is a hard purity failure. So the merges the search performs are only the ones it could not resolve, and splitting beyond the licence is explicitly not an error.

8 The algorithm assembled

Every ingredient is now defined, and it is worth seeing them composed before the refinement stage is added, because two structural facts are visible in the control flow and invisible in the prose.

One loop, not two. The obvious structure — cover the window to find candidates, then refine each candidate separately — is wrong, and the baselines measure how wrong. Any per-candidate refiner anchored on "the lowest point I can see" converges on one member of a close pair and leaves the other outside the interval it returns; this is exactly how grid-then-bisect falls from coverage 1.0 to 0.577. So cover and locate *alternate*, and they operate on *regions* rather than on candidate points. A region holding two eigenvalues then splits of its own accord, the moment a sample placed between them is excluded.

The gate is asked three times. It is consulted to decide whether a region may leave the cover loop (Algorithm 2, line 3), again after *every* locator step (Algorithm 3, line 9), and once more on the way out (line 14). Each catches a different failure, and all three are free: the gate is a pure function of samples already paid for.

That last point is worth stating as a rule, because it explains the shape of the whole algorithm.

Remark 12 (Only one line spends anything). In all three routines the sole operation that costs an evaluation is Evaluate. LiveRegions, DescentCertificate, Gate, LocatorStep and LooksLikeADip are pure functions of the sample arrays. This is why the design can afford to re-ask expensive-sounding questions at every step, and why the cost accounting in §10 counts cache *misses* and nothing else.

Algorithm 1: BRACKET — the top level

input : tension oracle $f(x) \mapsto (\sigma_1, \sigma_2, \dots)$; window $[a, b]$; precision prec
output : brackets in λ, with diagnostics — or a decline

work in $k = \sqrt{\lambda}$; $\quad \text{prec}_u \leftarrow \text{prec}/2$; // (6)
$(X, V, \text{settled}) \leftarrow \text{SEED}(f, a, b)$; // Weyl density, then nested doubling, §4
if $\neg$ *settled* **then return** DECLINE("nothing bounds the local slope");
$S \leftarrow \text{median}(V)$; // the plateau, measured once; Rem. 2
$L \leftarrow s \cdot \max_i |(V_{i+1} - V_i)/(X_{i+1} - X_i)|$; // (8), $s = 1.5$
budget $\leftarrow 600 \cdot \max\big(N(b) - N(a),\, 1\big)$; // per eigenvalue, never flat
floorLimited $\leftarrow$ FALSE;
for *round* $\leftarrow 1$ **to** 5 **do**
  $(X, V, L, \mathcal{R}, \text{stalled}) \leftarrow \text{COVER}(\dots)$; // Alg. 2
  floorLimited $\leftarrow$ floorLimited $\vee$ stalled;
  $(X, V, \mathcal{R}, \text{reopened}) \leftarrow \text{LOCATE}(\dots)$; // Alg. 3
  if *reopened* $= 0$ **then break**;
$\mathcal{K} \leftarrow \text{CONFIRM}(X, V, \mathcal{R}, L, S)$; // Alg. 3
return each $(a', b') \in \mathcal{K}$, widened by 4 ulp and mapped back by $u \mapsto u^2$, merging any that then abut ; // §3

Algorithm 2: COVER — refine by exclusion until every region is settled

output : samples, a possibly larger L, the live regions, and whether the floor bound
while true **do**
  $\mathcal{R} \leftarrow \text{LIVEREGIONS}(X, V, L)$; // cells with $\hat{s} < 0$, merged; (11)
  $\mathcal{T} \leftarrow \{\, R \in \mathcal{R} \;:\; \neg\,\text{PUREBYWIDTH}(R, \text{prec}_u) \;\wedge\; \neg\,\text{GATE}(R, L) \text{ certifies single} \,\}$;
   // silence is not permission
  if $\mathcal{T} = \varnothing$ **then return** $\mathcal{R}$, FALSE;
  if *no reduction in* $\sum_{\mathcal{T}} |R|$ *for 200 steps* ***or*** *budget spent* **then**
    return $\mathcal{R}$, TRUE ; // floor-limited, *not* a decline; §10.3
  $i \leftarrow \arg\min\{\, \hat{s}_i \;:\; \text{cells } i \text{ of } \mathcal{T} \,\}$; // the deepest unexplained dip
  $x \leftarrow \hat{x}_i$; $\quad$ EVALUATE f at x and insert into (X, V) ; // the Piyavskii point, (11)
  if x *was already a sample* **then return** $\mathcal{R}$, TRUE;
  if $s \cdot \max \textit{secant} > L$ **then**
    $L \leftarrow s \cdot \max \text{secant}$; $\quad$ growths += 1 ; // exclusions are recomputed next pass
    if *growths* > 16 **then return** DECLINE("the slope estimate keeps growing");

Algorithm 3: LOCATE, then CONFIRM

```
1  procedure LOCATE(X, V, R, prec_u, L):
2      foreach region (a_0, b_0) ∈ R do
3          if PUREBYWIDTH or ¬ GATE certifies single then keep (a_0, b_0); continue;
4          (a, b) ← (a_0, b_0);
5          repeat
6              x ← LOCATORSTEP(X, V, a, b) ;          // vertex of σ_1^2, then straddle; §7
7              if x is NONE or already sampled then break;
8              EVALUATE f at x; update (a, b) to the new certificate;
9              g ← GATE((a, b), L) ;                  // free, and the answer changes
10             if g.neighbour ≤ b_0 − a_0 then break  // a pair after all;
11             if g.single ∧ g.neighbour ≥ 4 (b − a) then break   // isolated; hand it on;
12         until at most 200 times, within budget;
13         g ← GATE((a, b), L);
14         if g is NONE or g.neighbour ≤ b_0 − a_0 then
15             keep (a_0, b_0);   reopened += 1 ;     // too wide beats being wrong
16         else
17             keep (a, b);

18 procedure CONFIRM(X, V, R, L, S):
19     foreach region (a, b) ∈ R do
20         repeat
21             EVALUATE f at the Piyavskii point of the deepest cell in (a, b) ;   // not the
                widest gap
22         until 3 times, or until a certificate exists;
23         keep (a, b) if DESCENTCERTIFICATE exists, or min V ≤ (1/2) max V within it, or
            min V ≤ (1/2) S ;                         // the last disjunct: a region inside a well reads flat
```

8.1 Reading the algorithm

Four lines carry most of the design, and each was arrived at by a measured failure.

- **The TODO filter** (Alg. 2, line 3) has two disjuncts because a region may leave the cover loop for two quite different reasons: it is narrow enough that it *cannot* be impure whatever it holds, or the gate has certified that it holds one eigenvalue. A region the gate cannot judge satisfies neither and stays.
- **The two non-decline exits** in Alg. 2 — stalling, and asking for a point that is already a sample — return the regions in hand with a flag rather than an error. They mean the objective's floor is binding, which is a fact about the pencil and not a failure of the search.
- **The reopen path** (Alg. 3, line 14) discards the locator's work and returns the *original* region. It is what keeps the coverage guarantee when the gate was wrong, and it is why the top level loops at all: the re-opened region goes back through COVER, where the locator's own samples now sit beside one member of the pair and make exclusion cheap.

- **The third disjunct in** CONFIRM exists because a region lying entirely *inside* a well has every sample on the floor, so its internal ratio is ≈ 1 and it reads as flat. Flatness at the floor is the opposite of flatness on the plateau, which is why the test is a fraction of the measured S (Remark 2).

Finally, note what is *not* in the pseudocode: any reasoning about whether two nearby eigenvalues may be merged. The search never decides that. It contracts until a region is pure by width or certified single, and the merges it performs are only the ones it could not resolve.

9 Refinement

The bracket is now handed to a refiner, which drives it to the requested accuracy.

9.1 Iterated parabolic interpolation, not Brent

Because (17) is exactly quadratic, the right algorithm solves for the vertex from three points and iterates. Measured over 26 eigenvalues at prec $= 10^{-14}$, on brackets of width 10^{-6} to 10^{-1}:

refiner	median relative error	evaluations per bracket
iterated parabolic on σ_1^2	$\mathbf{1.6 \times 10^{-16}}$	**5.5–7.9**
Brent on σ_1^2	5.4×10^{-10} to 5.6×10^{-5}	5.0–11.7
straddled interpolation on σ_1	0 (exact)	48–63

Brent does not exploit the parabolic structure: its parabolic step is guarded by a golden-section fallback, and it terminates long before the curvature information is used up. A general-purpose scalar minimizer is the wrong tool here.

Remark 13 (Squaring does not cost half the significant digits). The familiar $\sqrt{\varepsilon_{\mathrm{m}}}$ limit — that a minimum can be located only to the square root of machine epsilon — belongs to methods that locate a minimum by *comparing function values*, where the quadratic flatness of the objective destroys the comparison. Solving for the vertex from the *curvature* does not suffer it: the information is in the second difference, not in which of two nearly equal values is smaller.

9.2 Refine in λ

§3 argued for k on the strength of the wall slope. The curvature says the opposite, and it governs here. Fitting a quadratic to σ_1^2 over *identical physical sample points*, the residual relative to the data range is smaller in λ than in k in 30 of 32 levels across four rectangles, with a median advantage of 4.3× in residual and 4.8× in the accuracy of the located vertex (Figure 7). The originating measurement reported 17 of 17 and a median of 5.1×.

This deserves an honest note, because it contradicts the model. Writing the cone as $\sigma_1 \approx \sqrt{\varepsilon^2 + c^2(k - k^*)^2}$ makes σ_1^2 exactly quadratic *in* k; the data say it is closer to quadratic in λ. Fitting the model form directly leaves a residual of 1.5×10^{-2} against 8.7×10^{-4} for a plain quadratic in λ on the same points. So (2) is a good local model stated in a slightly wrong variable, and the two stages of the search are entitled to disagree about which variable they want because they are using different features of it.

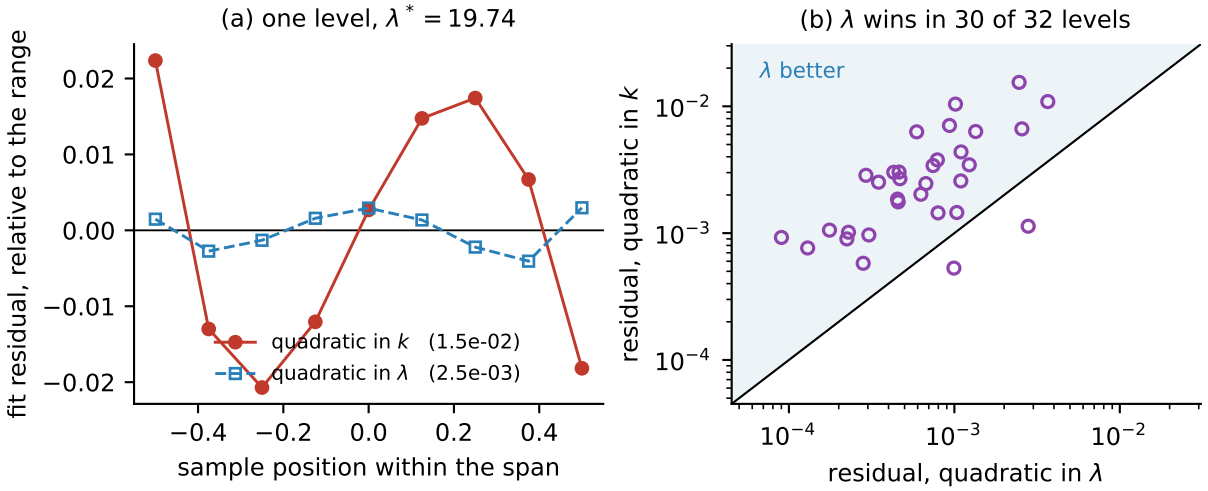


Figure 7: *(a)* Residuals of a quadratic fit to σ_1^2 at one level, over the same nine physical points, in k and in λ. The fit in k leaves a systematic W-shaped residual an order of magnitude larger. *(b)* The same comparison at every isolated level of four rectangles.

9.3 Tolerance, and the guards

The stopping tolerance is

$$x_{\text{atol}} \;=\; \max\bigl(\text{prec}\cdot|\lambda|\,,\; 4\,\text{ulp}(\lambda)\bigr). \tag{18}$$

The second term is not decoration: a tolerance derived from λ alone can *exceed* the bracket it is refining — a 2.7×10^{-13} bracket handed a tolerance of 7.9×10^{-13} — so the refiner terminates before doing anything and returns a point that is not the minimum.

Four guards, each named with the failure it prevents.

1. **Refuse when the parabola never closed.** A non-positive leading coefficient means the three points do not describe a single minimum: an empty bracket, or one holding two well-separated eigenvalues where the fit is across a W. In the second case the lowest sampled point can be a bracket *endpoint*, far from either eigenvalue, and would be returned as a success.

2. **But only when there was no progress.** The same condition also fires at the *end* of a good refinement, once the three points have collapsed onto the floor and the curvature is gone. Treating that as failure discards correct answers.

3. **Track the lowest point seen independently.** The triple is re-formed by position rather than by value, so a bad vertex can otherwise discard the sub-interval holding the minimum.

4. **Accept against the cone outside the bracket**, where it must be rising — not against the bracket's own ends, which are all near the floor once the bracket is narrow.

9.4 The interiority trap

The last guard has a subtlety that produced the same bug twice, in two forms, and it generalizes past this algorithm.

The naive version demands that an accepted minimum be strictly interior to its bracket. But a bracket is guaranteed to *contain* the eigenvalue, not to have the eigenvalue comfortably inside it: the cover loop stops on purity-by-width, and a well 10^{-7} wide can sit inside an 8.6×10^{-7} bracket with its minimum very near an end. An interior-minimum rule discarded 8 of 26 correct brackets.

The refined version of the same error is subtler. Probe interiority by evaluating at $x \pm h$ for a *fixed* relative step $h = 10^{-9}k$ and demanding both exceed $f(x)$. This demonstrates interiority only when the located root is already nearer the true minimum than h is; otherwise one step moves *toward* the minimum, reads lower, and the genuine root is refused. On an L-shape window this threw away all three eigenvalues at the default setting, every one of which had been located to at worst 1.2×10^{-8}.

The right way to state the finding is as a separation rather than a threshold. The kept and dropped populations split by a factor of eight in the ratio (location error)$/h$ — 0.000 to 0.098 against 0.782 to 2.974 — but *not* at 1, because the probe compares tension values on flanks of unequal slope. A first version of the corresponding test asserted error $> h$ exactly and was falsified by the row at 0.78; what survives is the mechanism and the separation, not a constant.

Remark 14. The lesson generalizes, and it is the one to take away from this section. A probe whose reach is set by the bracket width cannot know what it is stepping across. A probe whose reach is set by a *constant* cannot know that either, and additionally cannot know how accurate the point it is stepping from is. The acceptance test that survives compares the candidate against the two *sampled* endpoints it already has, and asks nothing new of the objective.

10 Cost, and the shape of the claim

10.1 Cost

Including the refinement each bracket hands on, at prec $= 10^{-8}$ on real pencils:

window	eigenvalues	evaluations per eigenvalue
a resolvable pair, $\lambda \in [10, 88]$	4	27
a multiplicity-3 cluster at $50\pi^2 \pm 3\%$	3	27
a wide window, $\lambda \in [10, 520]$	35	26

Roughly 26 evaluations per eigenvalue, near-flat in window size and in prec from 10^{-3} to 10^{-14}. For comparison, a uniform grid dense enough to detect everything costs about 43 per eigenvalue *and fails purity*, and grid-then-bisect costs 74–95 *and loses a third of the eigenvalues.*

10.2 The claim, stated conditionally

By Proposition 2 the completeness claim is not a theorem, and it must be reported as what it is:

> No x in $[a, b]$ with $\sigma_1(x) \le \tau$ exists outside the returned set, *given* the Lipschitz constant L.

Every run therefore reports L, the largest observed secant, the ratio between them, the number of times L grew after being relied upon, and the evaluation count. A large margin means the scan was conservative; a small one means the constant was being pushed and the run deserves a rerun at a larger safety factor. This matters because the failure mode is silent: a dip that is never sampled is invisible to any finite-evaluation method, and the exclusion search in particular returns nothing after its seed, with no error at all.

10.3 Declining is a third outcome

A method may assume the objective is well behaved at the scales its precision makes relevant, *detect that it is not*, and stop. Two triggers: the seed's slope estimate never settles (§4.3), or a long run of samples buys no reduction in the unresolved regions.

Declining is neither a pass nor a failure, and it is only honest if reported. A method that declines whenever it is about to be wrong has a perfect record over what it accepted and has demonstrated nothing, so the decline rate is quoted beside the pass rate over the same case set, always.

There is a related and easily conflated case which is *not* a decline: being asked for more precision than the objective can deliver. If the pencil's floor is binding, the right answer is to emit the brackets found, flag them floor-limited, and report the attained precision beside the requested one — not to report that nothing was found.

10.4 The band

The method's real limitation is best stated as a band. Write d for the separation of two eigenvalues at level x. Then

$$\frac{d}{x} < \text{prec} \quad \text{(merging is licensed)}, \qquad \frac{d}{x} > \text{the search's resolution} \quad \text{(must split, and can)},$$

and in between lies $\text{prec} < d/x < \text{resolution}$: *must split, and cannot.* The band's width is the ratio of the resolution to prec, so

> the harder the precision requested, the wider the band of separations that cannot be handled.

This is a property of the configuration rather than a tuning failure, and it is the reason the honesty machinery of this section is not decoration. A method whose limits widen as it is asked to do better has to be able to say so.

11 What changed in the port

The algorithm was designed and measured in a standalone repository against a single family — the $(1+\delta) \times 1$ rectangle with a plane-wave basis — and then ported into `chladni`. This section records the differences, because the gap between what was intended and what the implementation found is usually the most useful part.

11.1 What did not change

Almost all of the tuned constants, which is the more surprising result: safety factor 1.5, straddle 0.02, 1.5 samples per expected gap, a 10% settling tolerance, a 64-point seed without a Weyl hint, six doublings, the stall and locator caps at 200, the dip ratio 0.5, three confirmation evaluations, five cover/locate rounds, an isolation clearance of four bracket widths, and a gate margin of 1.5. The port is a set of *structural* corrections, not a retuning.

Remark 15. The gate margin deserves a footnote, because it invites a misreading: the originating module's own constant reads 2.0, but the searcher that uses it already defaulted to 1.5. The constant that ships is unchanged.

11.2 Genuine differences

	original	`chladni` v1, and why
architecture	an append-only `Evidence` store; every component a pure function of it, returning `NeedSample(x)` rather than evaluating	flat sample arrays with cover/locate/confirm in rounds, and a view object sharing the caller's cache at the $\lambda \leftrightarrow k$ seam. Offline testability traded for fitting the project's stage seams and a single shared, counted objective.
vertex	the parabolic vertex through the raw triple	solves for the quadratic coefficients by divided differences and *requires the leading coefficient to be positive*: a non-convex fit means the triple does not describe a single minimum.
fallback when the fit fails	sample the prediction, or the triple's midpoint	**bisect the wider side**. Using the straddle as a fallback shrinks the bracket by 2% per evaluation — measured creeping $6.2 \times 10^{-8} \to 3.4 \times 10^{-8}$ over twelve evaluations, then stopping.
straddle side	gaps to the nearest evidence samples	gaps to the certificate triple's own endpoints.
multiplicity reading in the gate	a parameter, defaulted to off	removed outright: a knob whose only safe setting is "off" is not a knob; see (16).
slope-growth cap	4, then decline	16. Growth is *cost, not error*, because exclusions are recomputed from the current L every pass. On a multiplicity-3 cluster the seed settles at 0.31 against a true wall slope of 0.683 and needs 7 growths to reach the right answer; at 4 it declined having found nothing, at 8 it passes at 80 evaluations.
exhaustion	abort	emit what is held, flag it floor-limited, and report the attained precision beside the requested one.
budget	an in-flight cap on a counted objective	600 evaluations *per expected eigenvalue*, never flat. A flat cap silently penalises dense windows, and once turned a 38-evaluation miss into the recorded claim that a whole family was unhandleable.

	original	`chladni` v1, and why
diagnostics	version-tagged benchmark tables	required output: L, its history, growths after use, the largest secant, attained against requested precision. The completeness claim is conditional on L, so L must be reported.
refiner start	—	the deepest *cached* sample rather than the arithmetic midpoint. A bracket is guaranteed to contain the eigenvalue, not to have a sampled descent inside it; this moved a located value from 1.8×10^{-10} to 7.4×10^{-12}.
refiner interior margin	—	removed (§9.4). It refused genuine off-centre minima — a level 8.8×10^{-8} from a bracket end against a demanded 2.15×10^{-7} — and was redundant with the test that the best value beat both ends.

11.3 Defects the port introduced, and what caught them

These are not differences of design; they are bugs the reimplementation created and the test tiers found. They are listed because each was found *by running, not by reading*, and because each is a plausible thing to do again.

- The budget defaulted to one eigenvalue where the objective offered no Weyl count, so the locator exhausted it and returned partial results *silently*.
- Regions were retired by overlap, so bracketing one member of a pair retired the whole region and lost the other — coverage 0.75 on every pair, which is §6's own failure mode arriving through the bookkeeping.
- The fast and slow paths were collapsed. The straddle places its sample at a predicted vertex, so on a region holding two wells it piles samples onto one and never lands between them: coverage 1.00 with purity failing at every precision. A region the gate has not certified must go back to the exclusion ladder.
- "Silence is not permission" applies on the way *out* of the locator too: with the post-check unable to certify, keeping the narrowed bracket discarded whatever else the region held.
- A region lying entirely *inside* a well reads as flat — every sample on the floor, internal ratio ≈ 1 — and was dropped by the confirmation step. Flatness at the floor is the opposite of flatness on the plateau, which is why the level test is a fraction of the measured plateau (Remark 2).
- The loop budget counted cache *misses*, so a refiner re-reading known points spun forever at zero cost.

11.4 Three stopping rules kept as comments

The implementation records three rules that were tried and removed, each with the number that removed it: a positive threshold on the envelope (broke 66 of 66 cases, Remark 5); a flat-bottom stop derived from $4\sigma_{\min}/c$, whose estimate is an *upper* bound on the floor and so stopped narrowing far too early; and a prominence stop on the certificate triple, which is sound in principle and unsafe early for the same reason the further-flank normalization exists. Recording a rejected idea beside the code, with its measurement, is what makes the design legible; it is also why the list in §11 could be written at all.

12 What is derived and what is measured

A reader who takes one thing from this note should take the difference between these two columns, because the algorithm's constants were all measured on *one* family and several still carry recalibration notes.

Follows from the model	**Measured, and may not transfer**
$p = 1/2$ is the unique power making the wall slope level-independent, (5)	that the spread in k is as small as 5–6%, and the value $C \approx 0.27$
the envelope and its minimizer, Lemma 1	that L from the largest secant is nearly tight
the ladder ratio $(s+1)/(s-1)$, independent of c, Proposition 3	safety $s = 1.5$, gate margin 1.5, straddle 0.02
σ_1^2 quadratic $\Rightarrow$ an exact vertex from three points	that λ beats k for the fit (30 of 32 here, 17 of 17 there)
the precision factor $\mathrm{prec}_u = p \cdot \mathrm{prec}$, (6)	$\sim$26 evaluations per eigenvalue
the cancellation-free c, (15)	that σ_2/c under-reads where it errs
$\sigma_2/\sigma_1 = 1 + d/u$, (16)	the $\lvert\Omega\rvert \,/\, \lvert\partial\Omega\rvert$ constant $\kappa \approx 2.6$

The originating design's own summary of its position is worth quoting, because it is the right posture toward every number above: the largest uncertainty is not in any of the individual measurements, but in the fact that all of them come from one rectangle. Several have since been re-measured on the disk, the equilateral triangle and the L-shape; the ones that have not are the ones to check first.

Sources

The design records are `docs/design/CHLADNI_V1.md` (the bracketing spec and its as-built deltas), `docs/design/search.md` (the precision contract, the ground-truth definition, and the multiplicity and collapse rules not developed here) and `docs/design/candidates.md` (the stage seams and the comparison protocol). The implementation is `tests/v1_bracket.py` and `tests/v1_refine.py`, which live in `tests/` because a scored candidate lives there until it wins. Figures are regenerated by `docs/figures/make_bracket_figures.py`, which drives the same code the text describes and prints every number quoted in a caption.

Not developed here, and specified in `search.md`: the multiplicity stage, which reads σ_{j+1}/σ_j against a wide bracket at zero additional cost; the collapse-versus-merge rule for roots reported close together; and the prec-indexed ground truth $S(\mathrm{prec})$ against which all of it is scored.